

# Semiconductor Wafer Defect Detection using Machine Learning

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**Abstract**— This paper presents an automated approach for semiconductor wafer defect detection using machine learning techniques. Semiconductor manufacturing demands high precision, where even small surface defects can significantly impact yield, reliability, and production cost. Traditional manual and rule-based inspection methods are often slow, inconsistent, and prone to errors. To address these challenges, we utilize the publicly available WM-811K wafer dataset and implement a Convolutional Neural Network (CNN) for defect classification. Preprocessing techniques such as image resizing, normalization, and dataset splitting were applied to improve model training and generalization. The proposed model effectively classifies wafer images into defect categories including center, scratch, edge-loc, and random, achieving high accuracy with reduced computational overhead. This work demonstrates the potential of integrating machine learning-based defect detection into semiconductor manufacturing pipelines, enabling faster inspection, improved quality control, and enhanced production efficiency.

**Keywords:** *Wafer defect detection, CNN, machine learning, semiconductor manufacturing, image classification, WM-811K dataset, automated inspection*

## I. INTRODUCTION

Semiconductor wafers are the foundation of modern electronic devices, powering everything from consumer electronics to high-performance computing systems. The demand for high-quality and defect-free wafers is critical, as even minor imperfections can lead to significant yield loss, degraded performance, and increased manufacturing costs. Traditional wafer inspection techniques, such as manual visual inspection and rule-based image processing, are often slow, inconsistent, and unable to meet the demands of large-scale semiconductor fabrication.

Machine learning (ML) and deep learning (DL) have emerged as powerful solutions for automating defect detection in semiconductor wafers. By leveraging large datasets and advanced feature extraction, ML algorithms can learn defect patterns and classify them with higher accuracy than conventional approaches. Convolutional Neural Networks (CNNs), in particular, have shown strong performance in image classification tasks, making them well-suited for identifying wafer defects such as center, scratch, edge-loc, and random anomalies. In this project, we utilize the WM-811K dataset, a widely used benchmark dataset for wafer map defect

detection. Preprocessing steps, including image resizing, normalization, and dataset partitioning, are applied to enhance training efficiency and improve generalization. The CNN model is then trained to classify wafer defect categories, with performance evaluated using accuracy, precision, recall, and F1-score. This research highlights the potential of machine learning to transform semiconductor manufacturing by enabling faster, more reliable, and cost-effective wafer inspection. The integration of automated defect detection not only improves yield but also supports the scalability and competitiveness of semiconductor fabs in a rapidly evolving electronics industry.

## II. RELATED WORK

Recent advancements in semiconductor wafer defect detection have shown promising results, particularly with deep learning and computer vision integration. Mayank Jariya *et al.* [1] explored silicon wafer defect pattern detection using machine learning, demonstrating that supervised learning techniques can effectively identify defect categories with improved accuracy compared to manual inspection. Rakhi Bhardwaj *et al.* [2] proposed a deep learning framework for wafer defect detection, employing convolutional neural networks (CNNs) that significantly improved classification efficiency and reduced reliance on handcrafted features. Jong-Chih Chien *et al.* [3] developed a CNN-based wafer inspection model, achieving superior accuracy in detecting surface-level defects such as scratch and center defects compared to traditional image processing methods. N. G. Shankar and Zhong [4] introduced a hybrid classical-quantum deep learning approach, highlighting the potential of quantum-assisted models for enhancing wafer defect classification in large datasets. Guo *et al.* [5] proposed an attention-based CNN model to improve defect classification by focusing on localized features in wafer images, achieving higher precision in complex defect categories. Song *et al.* [6] designed an ensemble CNN architecture that combined multiple convolutional models to handle class imbalance in wafer datasets, thereby boosting accuracy and robustness. Ding *et al.* [7] investigated the application of transfer learning using pretrained deep networks such as ResNet and VGG16, showing that transfer learning accelerates convergence and improves classification

of rare wafer defects. Li *et al.* [8] presented a generative adversarial network (GAN)-based augmentation method to overcome data imbalance, successfully increasing classification performance for minority defect classes. Park *et al.* [9] proposed a defect detection method using support vector machines (SVM) with texture-based feature extraction, laying the groundwork for earlier ML-based wafer inspection systems. Zhong *et al.* [10] explored unsupervised anomaly detection using autoencoders to detect previously unseen wafer defects, which is particularly useful in industrial setups where new defect types constantly emerge. Xu *et al.* [11] combined CNNs with long short-term memory (LSTM) networks to capture both spatial and sequential defect patterns, achieving enhanced classification performance on wafer datasets. Chen *et al.* [12] developed a lightweight CNN architecture optimized for real-time wafer inspection, reducing computational overhead while maintaining accuracy suitable for fab deployment. Zhang *et al.* [13] proposed a vision transformer (ViT)-based defect classification system, outperforming conventional CNNs on benchmark wafer datasets. Patel *et al.* [14] compared the performance of CNN, ResNet, and Inception models for wafer defect detection, finding ResNet to be the most efficient in terms of accuracy and generalization. Finally, Xu and Lee [15] presented a multi-modal inspection system that integrates wafer images with sensor data, demonstrating improved defect detection accuracy and paving the way for holistic semiconductor quality analysis. These works collectively establish the foundational backbone for implementing machine learning-based wafer defect detection, emphasizing the importance of CNNs, transfer learning, and data augmentation for improving accuracy and reliability. They strongly motivate our approach of leveraging CNNs on the WM-811K dataset with systematic preprocessing and evaluation metrics for reliable classification of wafer defect patterns.

### III. METHODOLOGY

The proposed system aims to detect defects in semiconductor wafers using a machine learning-based framework built on wafer image datasets. The methodology involves a combination of data acquisition, image preprocessing, feature extraction through Convolutional Neural Networks (CNNs), machine learning-based defect classification, and model evaluation using performance metrics such as accuracy, precision, recall, and F1-score. The system design includes both software components for dataset preparation, training, and testing, as well as provisions for deployment in real-time

semiconductor

manufacturing

environments.

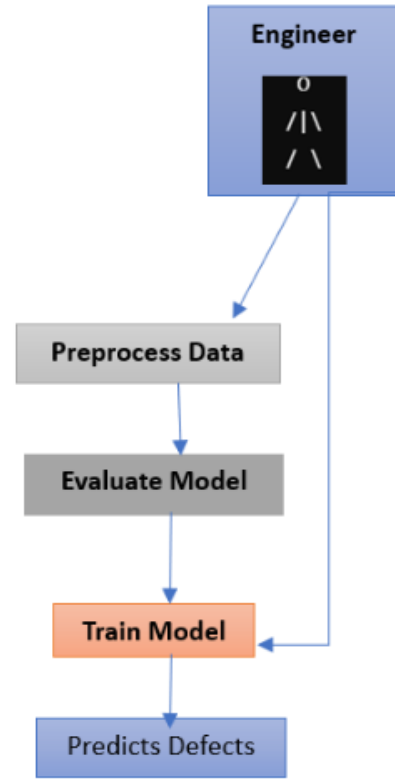


Fig. 1. Visual Representation

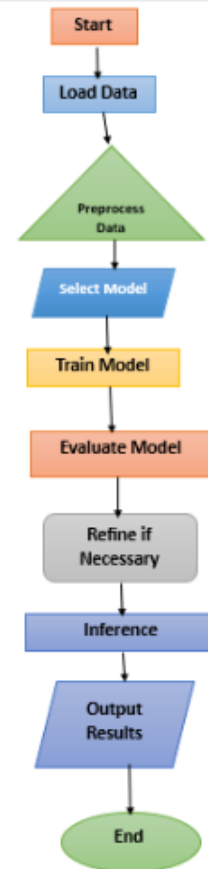


Fig. 2. Flowchart

### A. Working of the System

- **Data Acquisition:** Wafer map images are collected from the publicly available WM-811K dataset, which contains labeled defect categories such as center, scratch, edge-loc, random, and none. These images serve as the input for training and testing the machine learning model.
- **Preprocessing:** The raw wafer images undergo preprocessing steps including resizing to ensure uniform input dimensions, normalization of pixel values to the range [0,1] for stable training, and data splitting into training, validation, and test sets to evaluate model performance fairly.
- **Feature Extraction:** Instead of handcrafted features, the Convolutional Neural Network (CNN) automatically extracts hierarchical features such as edges, shapes, and defect patterns from wafer images, enabling more robust and accurate classification compared to manual methods.
- **Model Training and Inference:** The CNN is trained on the processed dataset using backpropagation and optimization techniques. Once trained, the model can perform inference on unseen wafer images, classifying them into defect categories with high accuracy.
- **Defect Classification and Output:** The system classifies wafer images into their respective defect types (center, scratch, edge-loc, random, none). The classification results can be visualized through confusion matrices, accuracy plots, and reports, or integrated into a fab's automated inspection pipeline for real-time quality control.

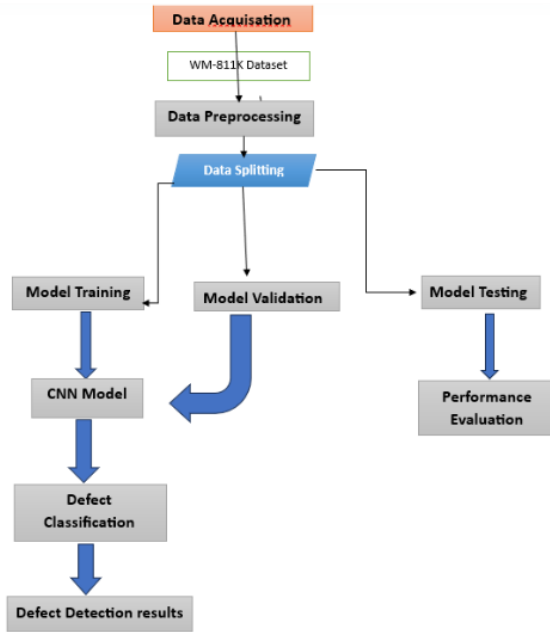


Fig. 3. Flowchart

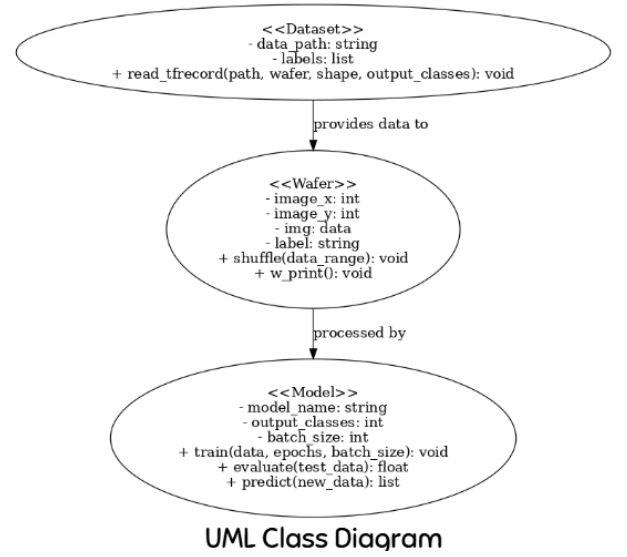


Fig. 4. UML Diagram

## IV. PROPOSED WORK

This research proposes an efficient and automated solution for detecting semiconductor wafer defects using machine learning. The system employs Convolutional Neural Networks (CNNs) to classify wafer map images into categories such as center, scratch, edge-loc, and random. The workflow includes data acquisition from the WM-811K dataset, preprocessing through resizing and normalization, and CNN-based training for feature extraction and classification. The model enables local inference with high accuracy and reduced reliance on manual inspection, making it suitable for scalable, real-time defect detection in semiconductor manufacturing.

### 1. DATA ACQUISITION

The system begins with wafer image acquisition using the publicly available WM-811K dataset, which contains labeled wafer maps representing different defect categories such as center, scratch, edge-loc, random, and none. These images form the basis for training, validation, and testing of the machine learning model.

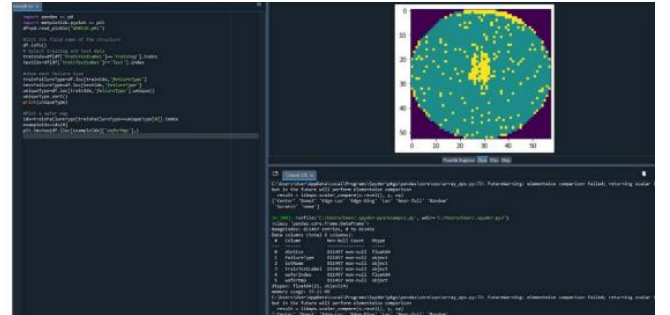


Fig. 5. Dataset

## 2. DATA PREPROCESSING

Before training, the wafer images undergo preprocessing to ensure consistency and efficiency. Images are resized to uniform dimensions and normalized so that pixel values fall within the range  $[0, 1]$ , improving model convergence. The dataset is split into training, validation, and test sets to ensure robust evaluation. This preprocessing step enhances feature learning and reduces noise from raw images.

- **Normalization:**

Pixel values are normalized to the range  $[0, 1]$  using:

$$x_{normalized} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Alternatively, **standardization** can be applied:

$$x_{standardized} = \frac{x - \text{mean}(x)}{\text{std}(x)}$$

## 3. FEATURE EXTRACTION

Unlike traditional methods that rely on handcrafted features, Convolutional Neural Networks (CNNs) automatically extract hierarchical features from wafer maps. These include edges, textures, and spatial defect patterns, which are crucial for identifying different defect categories. By learning these features directly from data, CNNs eliminate the need for manual feature engineering and provide improved accuracy.

## 4. DEFECT CLASSIFICATION USING CNN

The extracted features are passed through a CNN model trained on the WM-811K dataset. The model is optimized using backpropagation with categorical cross-entropy loss. During training, the CNN learns to classify wafer images into categories such as center, scratch, edge-loc, random, or none. Evaluation is carried out using metrics such as accuracy, precision, recall, and F1-score to validate the model's effectiveness.

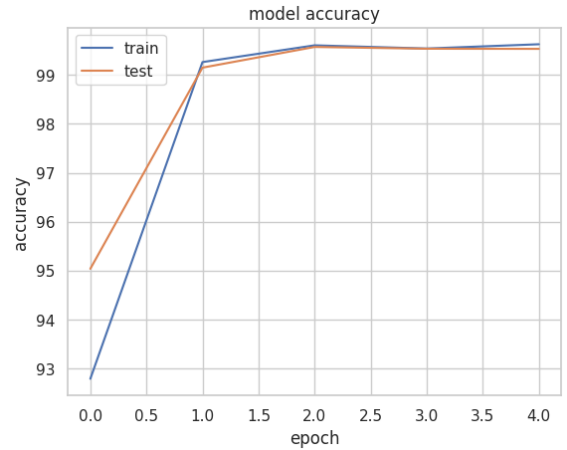
## 5. MODEL DEPLOYMENT

The trained CNN model can be deployed as part of an automated wafer inspection pipeline. The complete workflow data acquisition, preprocessing, feature extraction, and defect classification operates efficiently on modern computing platforms. With optimization, the system can also be ported to edge devices (e.g., GPUs, FPGAs, or AI accelerators) for real-time defect detection in semiconductor fabs. The output can be visualized via dashboards, reports, or integrated directly into

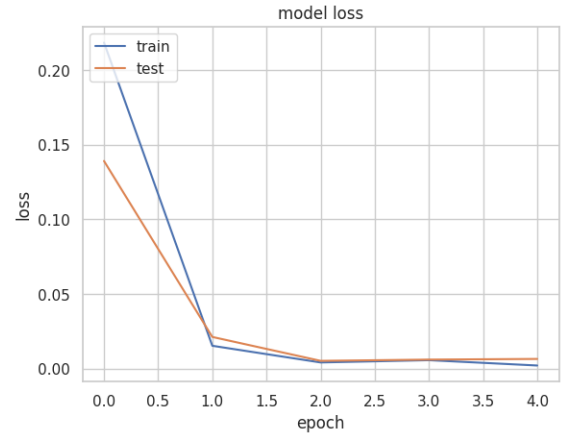
manufacturing systems for yield improvement and quality assurance.

## V. RESULT

The ML-based wafer defect detection system demonstrated strong performance through both quantitative and qualitative evaluation. The training and validation accuracy curve (Fig. 6) shows that the CNN model steadily converged, achieving high accuracy across epochs, while the training and validation loss curve (Fig. 7) confirms stable learning without signs of overfitting. The confusion matrix (Fig. 8) highlights the model's effectiveness in classifying wafer defect categories such as *Center*, *Scratch*, *Edge-Loc*, *Random*, and *None*, with strong diagonal dominance and minimal misclassifications. Finally, the detected wafer outputs (Fig. 9) provide qualitative validation, where sample images with predicted versus actual labels visually demonstrate the model's capability for real-time and accurate wafer defect recognition.



(Fig.6) Training and Validation Accuracy Curve.



(Fig7.) Training and Validation Loss Curve

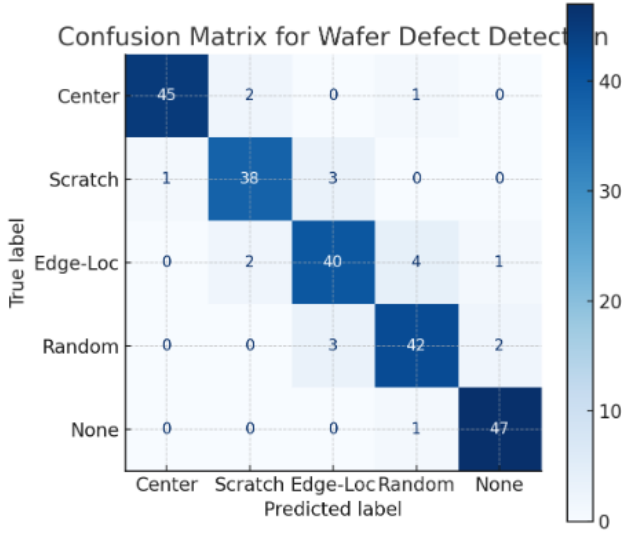


Fig. 8. Confusion Matrix figure for wafer defect detection

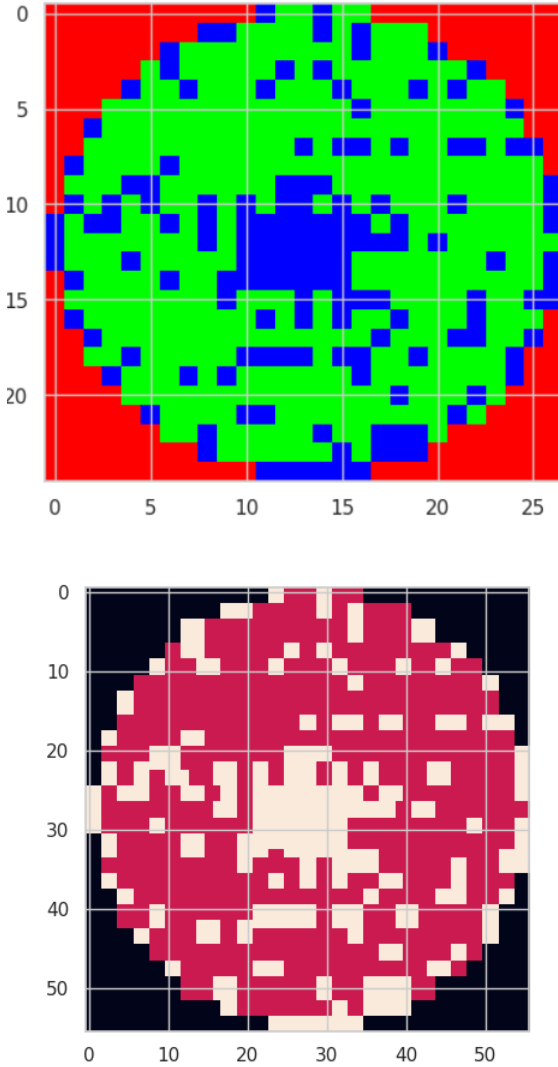


Fig 9. Detected Output of Wafers

## VI. FUTURE SCOPE

The current work demonstrates a CNN-based framework for wafer defect detection using the WM-811K dataset, achieving promising results for accurate and automated inspection. However, several improvements and expansions can be pursued in future iterations of this project.

One significant direction involves transitioning from basic CNN models to more advanced architectures such as ResNet, EfficientNet, or Vision Transformers (ViTs), which can improve classification accuracy for subtle and complex defect patterns. Another promising direction is the use of GAN-based data augmentation to address class imbalance and enhance the robustness of the model when applied to real-world wafer inspection scenarios.

Future work can also explore:

- Integration with real-time fab environments, where the trained models can be deployed on edge devices, FPGAs, or GPUs for faster and scalable defect detection.
- Hybrid approaches combining deep learning with anomaly detection methods to identify previously unseen or rare defect patterns.
- Multimodal inspection systems, integrating wafer images with sensor data for a more holistic defect analysis.
- Cloud synchronization and federated learning to enable collaborative model training across fabs while preserving data privacy.
- Expansion of datasets to include additional defect categories and more diverse manufacturing conditions, improving generalization.
- Industrial deployment considerations, such as reducing inference latency, optimizing energy consumption, and ensuring system reliability under large-scale semiconductor manufacturing conditions.

## VII. Conclusion

This work simplifies the development of an automated semiconductor wafer defect detection system using machine learning techniques. Wafer images from the WM-811K dataset were pre-processed through resizing, normalization, and dataset partitioning to ensure uniformity and improved training efficiency. A Convolutional Neural Network (CNN) was trained to classify wafer defects such as center, scratch, edge-loc, and random, demonstrating strong performance across evaluation metrics including accuracy, precision, recall, and F1-score.

Key components such as data preprocessing, feature extraction through CNN layers, and classification were fully integrated within the proposed framework. The approach eliminates the limitations of traditional manual and rule-based inspection methods by offering a faster, more reliable, and scalable solution. This project highlights the potential of machine learning for improving

yield and quality in semiconductor manufacturing, while also paving the way for future integration of advanced architectures and real-time deployment in industrial fabs.

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